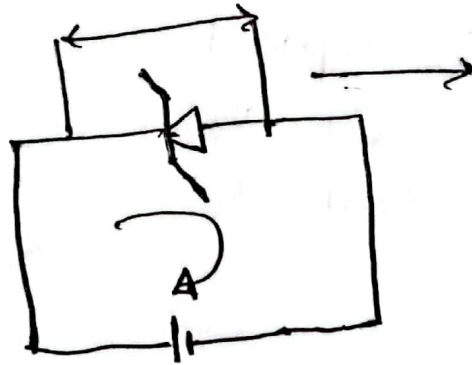


Zener Diode

Zener diode is a heavily doped semiconductor diode which is designed to conduct current in reverse bias and regulate voltage across it.

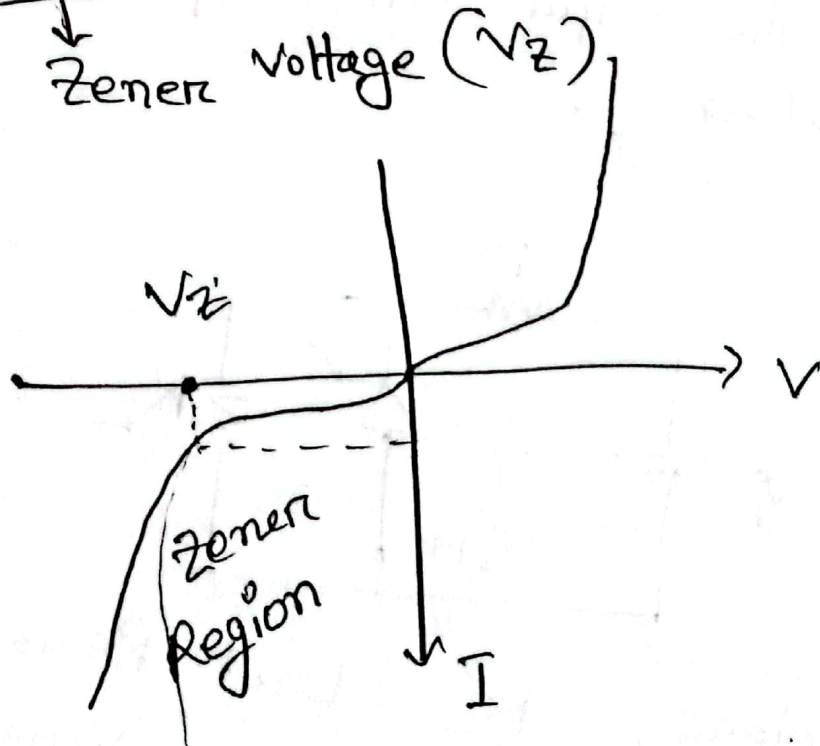
In Forward bias zener diode acts like P-N junction diode



voltage across this always remain constant

→ Reverse bias में connect कराने से current conduct करके नियम बना

Must निर्दिष्ट voltage $\geq$ input



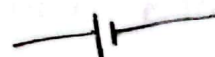
In case of Reverse
2 conditions

Applied voltage < Zener Voltage



diode will be replaced with open circuit as no current will be conducted

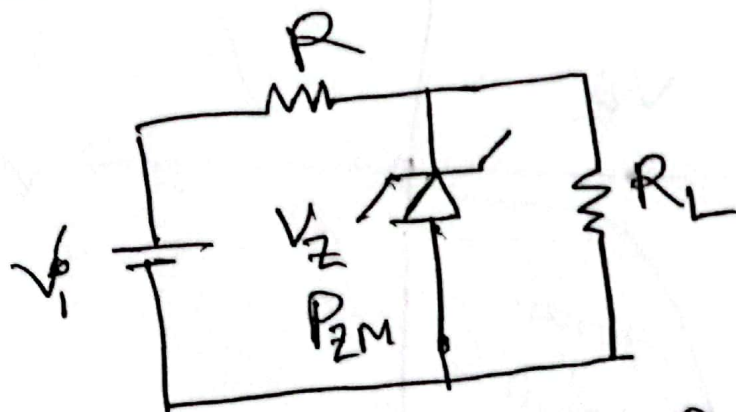
Applied > Z. Vol



diode will on
~~V_{applied}~~

Voltage Regulator

A combination of elements designed to ensure that the output voltage of a supply remains constant.



Here, R_L = Load Resistance

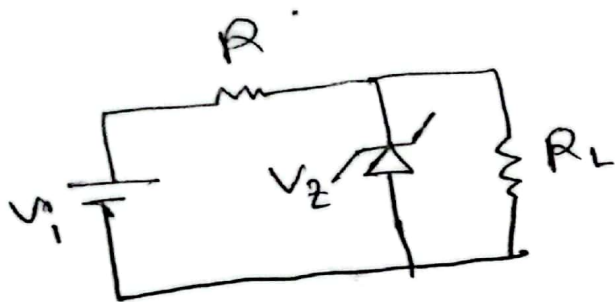
V_Z = Zener Voltage

P_{ZM} = Maximum Power dissipation of diode

V_i = Input voltage,

4 case

- V_i and R_L Fixed
- V_i fixed R_L variable
- V_i variable R_L fixed
- V_i, R_L Variable

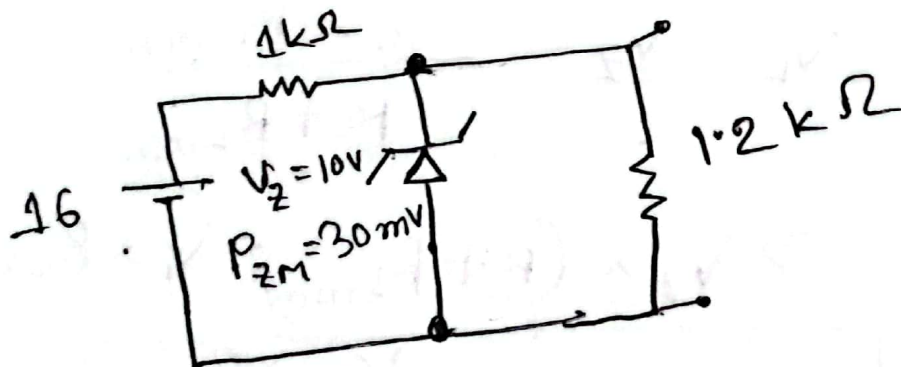


if $V_L \geq V_Z \rightarrow \text{ON}$
else OFF

Using voltage divider Rule,

$$V_L = \frac{R_L}{R + R_L} \times V_i$$

Ex-1 Determine V_L, V_R, I_Z, P_Z .



$$V_L = \frac{1.2 \text{ k}}{(1.2 + 1) \text{ k}} \times 16$$

$$= 8.72 \text{ V}$$

$V_L < V_Z \rightarrow \text{OFF} \rightarrow$ so Z.D replace with open circuit.

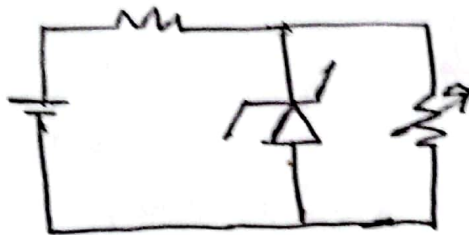
$$I_R = \frac{16}{1+1.2}$$

$$= 7.27 \text{ mA}$$

$$I_R = I_Z + I_L$$

$$\Rightarrow I_R = I_L$$

Type-2



1) Condition for Minimum Load Resistance

Voltage across Z-D = V_Z

$$\Rightarrow V_L = V_Z = \frac{V_i \times R_{L\min}}{R + R_{L\min}}$$

$$\Rightarrow V_Z \times (R + R_{L\min}) = V_i \times R_{L\min}$$

$$\Rightarrow R_{L\min} = \frac{V_Z \times R}{V_i - V_Z}$$

$$\therefore I_{L\max} = \frac{V_L}{R_{L\min}} = \frac{V_Z}{R_{L\min}}$$

$$\text{then, } I_R = I_{Z\min} + I_{L\max} \Rightarrow I_{Z\min} = I_R - I_{L\max}$$

$$\therefore I_{Z\min} = \frac{V_i - V_Z}{R} - I_{L\max}$$

ii) Condition for max R_L

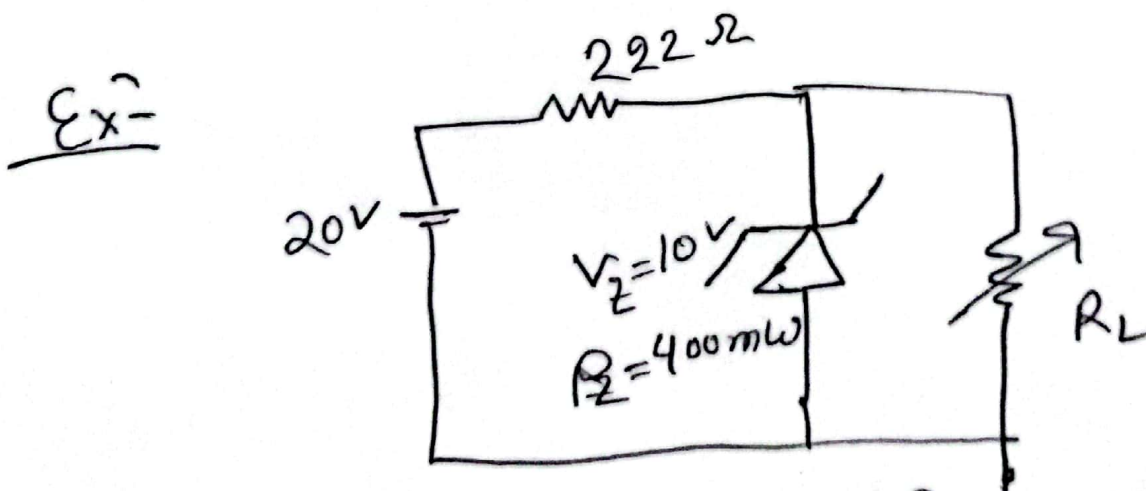
$$I_{L\min} \rightarrow R_{L\max} \rightarrow I_{Z\max}$$

$$P_{Zm} = I_{Zm} \times V_Z \Rightarrow I_{Zm} = \frac{P_{Zm}}{V_Z}$$

$$I_{L\min} = I_R - I_{Z\max}$$

$$I_{L\min} = \frac{V_i - V_Z}{R} - \frac{P_{Zm}}{V_Z}$$

$$\therefore R_{L\max} = \frac{V_L}{I_{L\min}} = \frac{V_Z}{I_{L\min}}$$



Ranges of $R_L = ?$

$$R_{L\min} = \frac{V_Z \times R}{V_i - V_Z} = \frac{10 \times 222}{20 - 10} = 222 \Omega$$

$$\therefore I_{L\max} = \frac{V_Z}{R_{L\min}} = \frac{10}{222} \approx 45$$

$$I_R = \frac{V_i - V_Z}{R} = \frac{20 - 10}{222} = 45 \text{ mA}$$

$$I_{L\min} = I_R - I_{Z\max} = 45 - 40 = 5 \text{ mA}$$

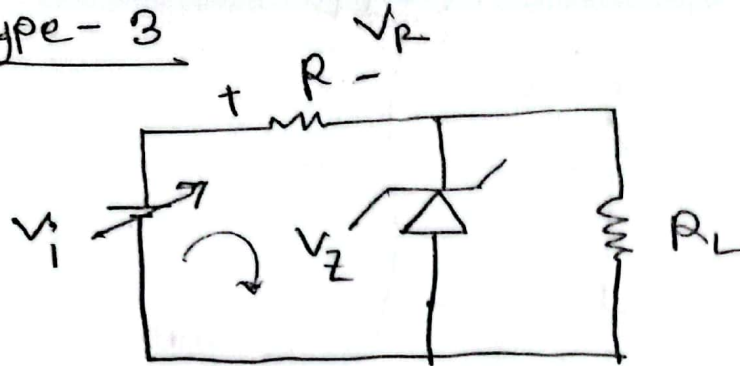
$$P_{Z\max} = 400 \text{ mW}$$

$$\therefore I_{Z\max} = \frac{P_{Z\max}}{V_Z}$$

$$= 40 \text{ mA}$$

$$R_{L\max} = \frac{V_Z}{I_{L\min}} = \frac{10}{5 \text{ mA}} = 2 \text{ k}\Omega$$

Type-3



1) Condition for $V_{i\min}$:

$$V_L = V_Z$$
$$\Rightarrow \frac{V_i \times R_L}{R + R_L} = V_Z$$

$$\therefore V_{i\min} = \frac{V_Z \times (R + R_L)}{R_L}$$

2) Condition for $V_{i\max}$:

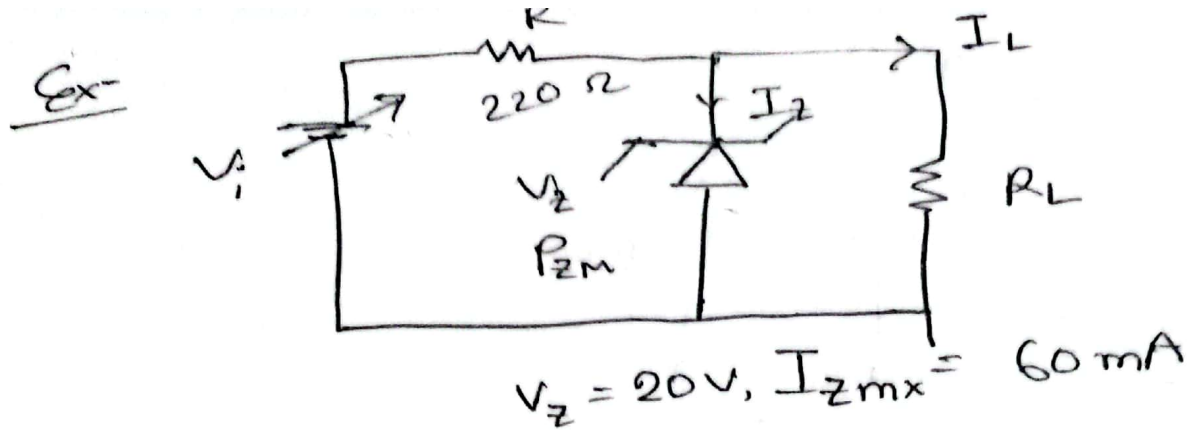
$V_{i\max}$ is limited by $I_{Z\max}$

$$I_{R\max} = I_{Z\max} + I_Z \rightarrow I_L = \frac{V_L}{R_L} = \frac{V_Z}{R_L}$$

$$V_{i\max} - V_{R\max} - V_Z = 0$$

$$\Rightarrow V_{i\max} = V_{R\max} + V_Z$$

$$V_{i\max} = I_{R\max} \times R + V_Z$$



Determine the range of input voltage (V_i) that will maintain Zener diode in ON state.

$$V_{imin} = \frac{V_Z \times (R + R_L)}{R_L}$$

$$= \frac{20 \times (220 + 1200)}{1200}$$

$$= 23.67 V$$

$$V_{imax} = V_{Rmx} + V_Z$$

$$= I_{Rmx} \times R + V_Z \quad \text{--- (1)}$$

$$I_{Rmx} = I_{Zmx} + I_L$$

$$= 60 + \frac{V_Z}{R_L}$$

$$V_i = I_{Rmx} \times R + V_Z$$

$$= 76.67 \times 220 + 20$$

$$= 36.86 V$$

$$I_L = \frac{V_Z}{R_L} = \frac{20}{1200}$$

$$= 60 + \frac{20}{1200}$$

$$= 76.67 mA$$